ECO 362: Financial Investments
Stocks

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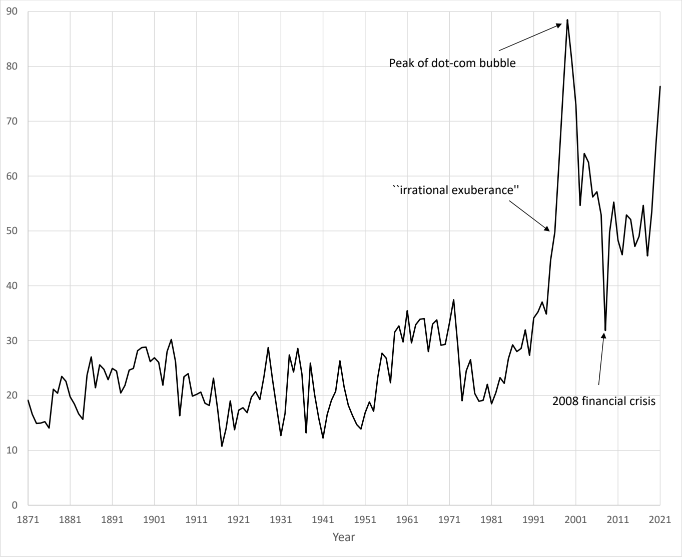
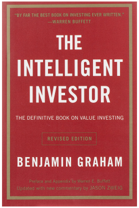
S&P composite index

- Founded in 1860, a value-weighted portfolio of large and liquid US stocks.
- Became the S&P 500 index in 1957.
- Price-dividend ratio: Current price index to dividends over the prior year.
  - Its inverse (i.e., dividend-price ratio) is often referred to as the dividend yield.

Price-dividend ratio for the S&P composite index



Contrarian investing



Benjamin Graham

"Buy when most people, including experts, are pessimistic, and sell when they are actively optimistic."

John Templeton

"Buy when others are despondently selling and to sell when others are euphorically buying takes the greatest courage, but provides the greatest profit."

Warren Buffett

"Be fearful when others are greedy, and greedy when others are fearful."

Interpreting the price-dividend ratio

1. Mean reversion: Tendency to rise when low and fall when high.
2. Two distinct periods.
    - Mean of 23 through 1981.
    - Higher mean of 47 since then.

- "Buy low and sell high."
    - Go into the stock market when the price-dividend ratio is low.
    - Sell out of the stock market when the price-dividend ratio is high.
- Is the price-dividend ratio permanently higher since late 1980s?

A probability refresher

- A random variable X_{t+1} at future date $t+1$.
- Its expectation at date t is $\mathbb{E}_t[X_{t+1}]$.
- The law of iterated expectations says that

$$\mathbb{E}_0[\mathbb{E}_t[X_{t+1}]] = \mathbb{E}_0[X_{t+1}]$$

- Rather than a formal proof, let's do an example.

An example of the law of iterated expectations

- Flip a coin today.
  - If heads, flip a coin again tomorrow and receive \$16 if heads (and \$0 if tails).
  - If tails, flip a coin again tomorrow and receive \$4 if heads (and \$0 if tails).
- Two ways to compute the expected value today.
  1. Use the formula $\mathbb{E}_0[\mathbb{E}_1[X_2]]$.

$$0.5(\underbrace{0.5 \times \$16 + 0.5 \times \$0}_{\text{heads today}}) + 0.5(\underbrace{0.5 \times \$4 + 0.5 \times \$0}_{\text{tails today}})$$

$$= 0.5 \times \$8 + 0.5 \times \$2 = \$5$$

  2. Use the formula $\mathbb{E}_0[X_2]$.

$$\underbrace{0.25 \times \$16}_{\text{heads-heads}} + \underbrace{0.25 \times \$0}_{\text{heads-tails}} + \underbrace{0.25 \times \$4}_{\text{tails-heads}} + \underbrace{0.25 \times \$0}_{\text{tails-tails}}$$

$$= \$4 + \$1 = \$5$$

A math refresher

- Let $0 < x < 1$ be a constant.
- A geometric sum is

$$\begin{aligned}\sum_{i=0}^{\infty} x^i =& 1 + x + x^2 + x^3 + \cdots \\ =& \frac{1}{1-x}\end{aligned}$$

- If we start the sum from $i = 1$,

$$\sum_{i=1}^{\infty} x^i = \frac{1}{1-x} - 1 = \frac{x}{1-x}$$

Dividend discount model

- Definition of return in year $t+1$:

$$1 + R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

- Assume constant expected returns.

$$\mathbb{E}_t[1 + R_{t+1}] = 1 + \overline{R} = \frac{\mathbb{E}_t[P_{t+1} + D_{t+1}]}{P_t}$$

- Rewrite as

$$P_t = \frac{\mathbb{E}_t[D_{t+1}]}{1 + \overline{R}} + \frac{\mathbb{E}_t[P_{t+1}]}{1 + \overline{R}}$$

- Start at $t = 0$ and substitute out future price recursively.

$$
\begin{aligned}
P_0 &= \frac{\mathbb{E}_0[D_1]}{1+\overline{R}} + \frac{\mathbb{E}_0[P_1]}{1+\overline{R}} \\
&= \frac{\mathbb{E}_0[D_1]}{1+\overline{R}} + \frac{1}{1+\overline{R}}\mathbb{E}_0\left[\frac{\mathbb{E}_1[D_2]}{1+\overline{R}} + \frac{\mathbb{E}_1[P_2]}{1+\overline{R}}\right] \\
&= \frac{\mathbb{E}_0[D_1]}{1+\overline{R}} + \frac{\mathbb{E}_0[D_2]}{(1+\overline{R})^2} + \frac{\mathbb{E}_0[P_2]}{(1+\overline{R})^2} \\
&= \cdots
\end{aligned}
$$

- The dividend discount model is

$$
P_0 = \sum_{t=1}^{\infty} \frac{\mathbb{E}_0[D_t]}{(1+\overline{R})^t}
$$

Gordon growth model

- Assume that dividend growth is a constant $G < \overline{R}$ per year.

$$\mathbb{E}_0\left[\frac{D_t}{D_0}\right] = (1+G)^t$$

- Write the dividend discount model as

$$\begin{aligned}\frac{P_0}{D_0} &= \sum_{t=1}^{\infty} \frac{\mathbb{E}_0[D_t/D_0]}{(1+\overline{R})^t} \\ &= \sum_{t=1}^{\infty} \frac{(1+G)^t}{(1+\overline{R})^t} = \sum_{t=1}^{\infty} \left(\frac{1+G}{1+\overline{R}}\right)^t \\ &= \frac{1+G}{1+\overline{R}}\left(1 - \frac{1+G}{1+\overline{R}}\right)^{-1} = \frac{1+G}{\overline{R}-G}\end{aligned}$$

Interpreting the Gordon growth model

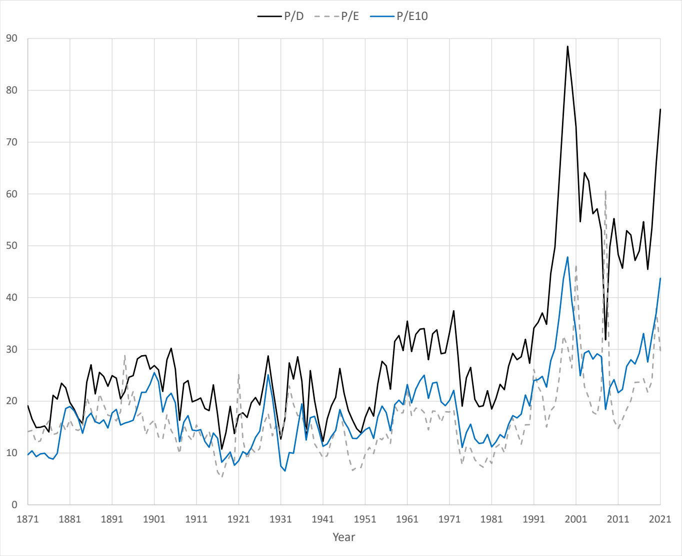
$$\frac{P_0}{D_0} = \frac{1 + G}{\overline{R} - G}$$

- When the price-dividend ratio is high, either
  1. Expected returns $\overline{R}$ are low: Forecasts low future returns.
  2. Expected dividend growth G is high: Forecasts high future dividend growth.
- Empirically, 1 is true. That is, a high price-dividend ratio forecasts low future returns.
- A high price-dividend ratio is a warning that a correction may be ahead (e.g., the dot-com bubble).
- A low price-dividend ratio signals a buying opportunity (e.g., the 2008 financial crisis).

Why is the price-dividend ratio higher since 1980s?

1. Firms switched from cash dividends to share repurchases (which are tax advantaged) after SEC Rule 10b-18 in 1982.
   - Dividends may no longer be a good signal of fundamental value.
   - Need to look at earnings or adjusted dividends that incorporate share repurchases.
2. Investors are less risk averse, so willing to hold the stock market at lower expected returns.
   - Recall the lecture on "CAPM": $\mathbb{E}[R_m - R_b] = \overline{\gamma}\mathrm{Var}(R_m)$.

Price-earnings ratio for the S&P composite index



S&P composite index (1920–2021)

Statistic	T-bills	Stocks	Stocks−T-bills
Mean	3.4%	12.3%	9.0%
Standard deviation	3.0%	19.5%	19.7%

- Equity premium: Average excess returns of stocks relative to T-bills is 9.0% annually.

Testing return predictability

- Write stock returns as

$$R_{s,t+1} = \underbrace{R_{b,t+1}}_{\text{riskless return}} + \underbrace{R_{s,t+1} - R_{b,t+1}}_{\text{excess return}}$$

- The riskless return is forecastable, but not obvious that the excess return is.
- Test whether the price-dividend ratio forecasts future excess returns through a time-series regression.

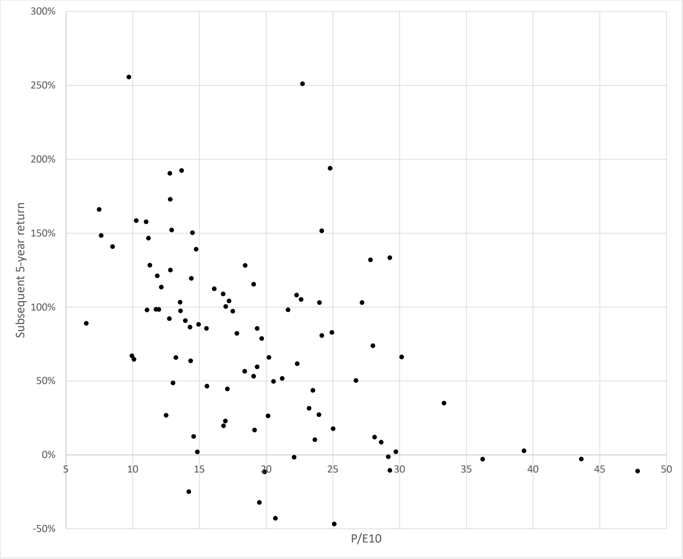
$$R_{s,t+1} - R_{b,t+1} = \alpha + \beta X_t + \epsilon_{t+1}$$

- Null hypothesis ($\beta = 0$): Price-dividend ratio does not forecast future returns.
- Alternative ($\beta < 0$): Price-dividend ratio forecasts future returns.

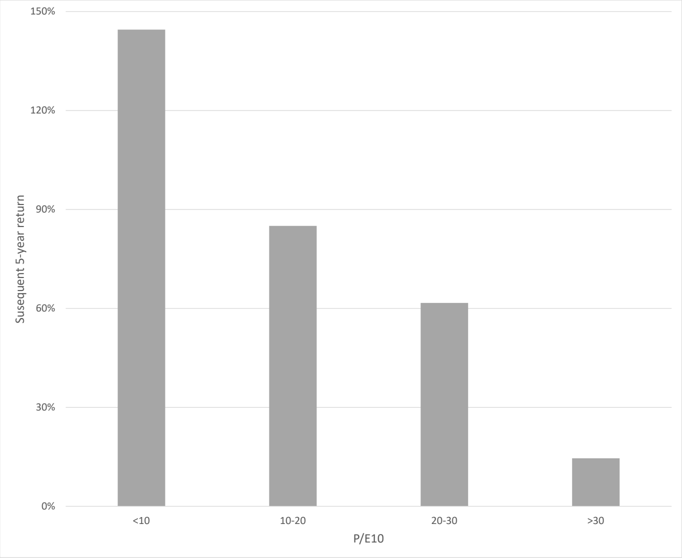
Example 1: Estimated regression

Variable	Sample	Estimate	*t*-stat
P/D	1920–2020	−0.002	−1.59
	1920–1981	−0.011	−2.63
	1982–2020	−0.003	−1.91
P/E10	1920–2020	−0.005	−2.24

P-E10 vs. subsequent 5-year returns



Average 5-year returns by P-E10



Example 2: Implications for portfolio choice

- Recall the optimal portfolio weight on stocks from the lecture on "Risk and Return".

$$w = \frac{\mathbb{E}[R_s - R_b]}{\gamma\sigma^2}$$

- Historically, $\mathbb{E}[R_s - R_b] = 9.0\%$ and $\sigma = 19.7\%$.
- For an investor with risk aversion $\gamma = 5$,

$$w = \frac{9.0\%}{5(19.7\%)^2} = 0.47$$

Market-timing strategy

- However, we just learned that expected excess returns are time varying.

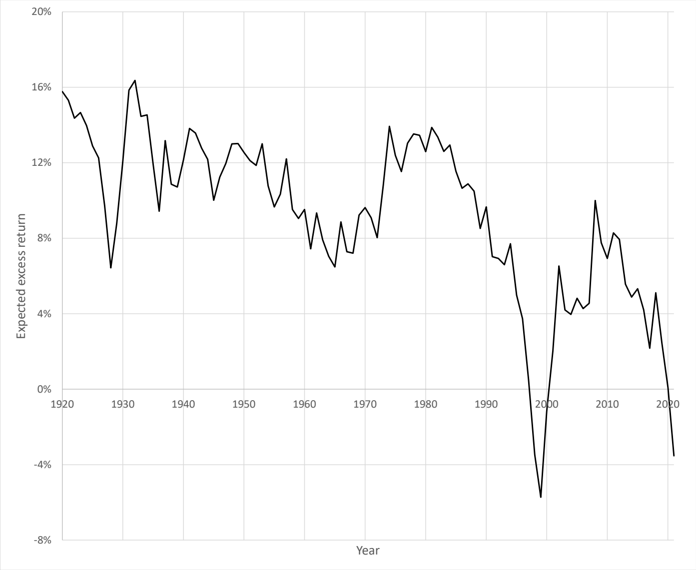
$$\mathbb{E}_t[R_{s,t+1} - R_{b,t+1}] = \alpha + \beta X_t$$

- So the optimal portfolio weight should also be time-varying.

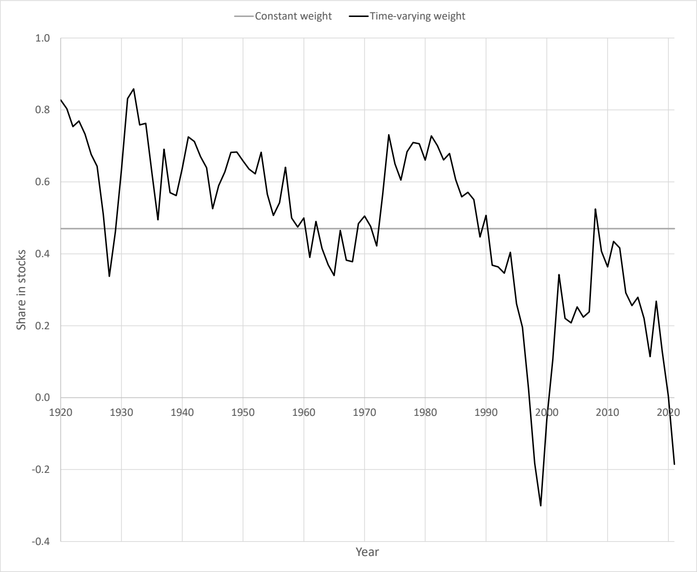
$$w_t = \frac{\mathbb{E}_t[R_{s,t+1} - R_{b,t+1}]}{\gamma\sigma^2} = \frac{\alpha + \beta X_t}{\gamma\sigma^2}$$

- Since $\beta < 0$, you must "buy low and sell high."
  - Increase share in stocks when the price-earnings ratio is low.
  - Decrease share in stocks when the price-earnings ratio is high.

Expected excess return

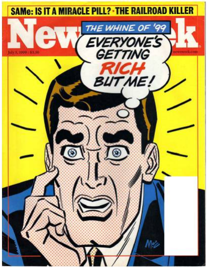


Optimal portfolio weight on stocks



Shorting the stock market in the late 1990s?

- Greenspan warned of "irrational exuberance" in a televised speech on 12/5/1996.
- "Crash, dammit" (*The Economist*, 10/16/1997).
- But most investors had optimistic views...



Newsweek
7/5/1999

Time
9/27/1999

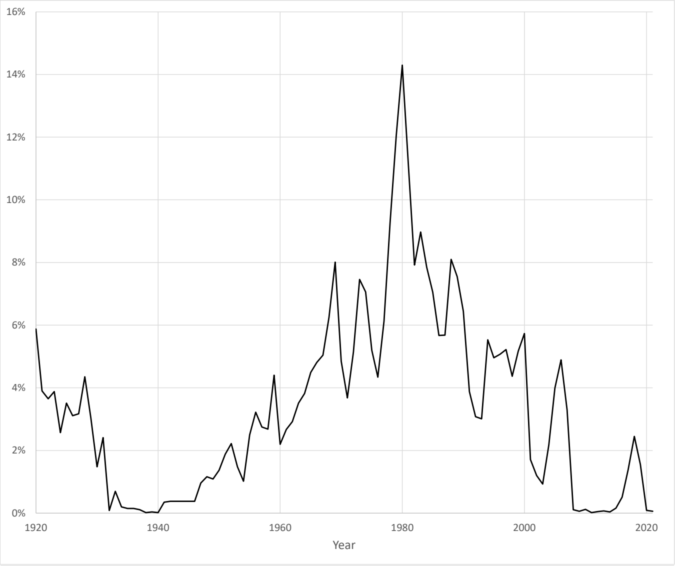
History of economic thought

Robert Shiller (1946–)
Nobel Prize in 2013 for return predictability and behavioral finance.

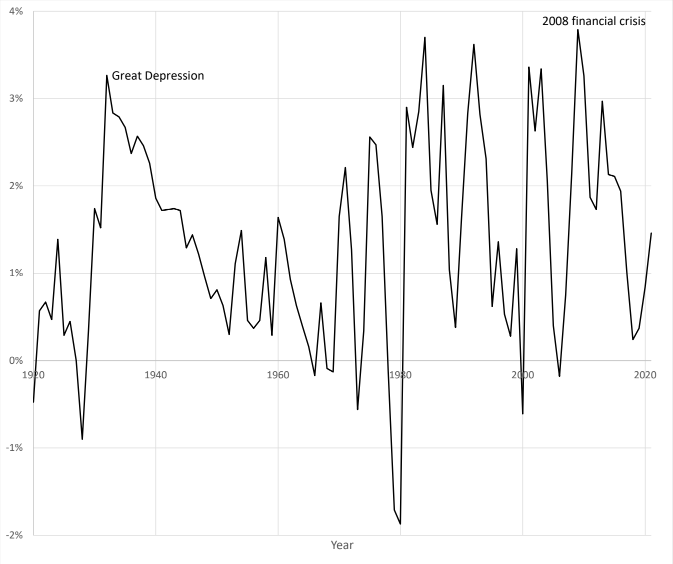
- Published "Irrational Exuberance" (March 2000), predicting the dot-com crash.
- Published "Is There a Bubble in the Housing Market?" (2003, *Brookings Papers*).

  So a very plausible scenario is that home-price increases continue for a couple more years, and then we might have a recession and they continue down into negative territory and languish for a decade. (Interview with NYT in August 2005)

3-month T-bill rate



Yield spread (10-year minus 3-month)



Example 1 (continued): Estimated regression

Variable	Sample	Estimate	t-stat
T-bill rate	1920–2020	-1.10	-1.67
Yield spread	1920–2020	2.20	1.34

- T-bill rate: A high nominal interest rate is related to high inflation, which is bad news for stocks.
- Yield spread: Yield curve steepens in recessions when prices of risky assets fall, including long-term bonds and stocks.

Applying the Gordon growth model to individual stocks

- The dividend-price ratio for stock i according to the Gordon growth model.

$$\frac{D_i}{P_i} = \frac{\overline{R}_i - G_i}{1 + G_i}$$

- Stocks that are trading at high dividend-price ratios are expected to have
  1. High future returns.
  2. Low future dividend growth.
- Empirically, 1 is true.
- Value strategy: Buy stocks with high dividend-price ratios (and short stocks with low dividend-price ratios).

Various flavors of value

1. Dividend-price ratio (a.k.a. dividend yield).
    - Not all firms pay dividends.
2. Earnings-price ratio (a.k.a. earnings yield).
    - Younger firms may not have earnings yet.
3. Book-to-market equity (preferred measure among academics).
    - Book equity is a measure of shareholders' equity, based on historical cost accounting.

- Ratios of "fundamentals" to market value.
    - Value stocks have high dividend-price, earnings-price, or book-to-market equity.
    - Growth stocks have low dividend-price, earnings-price, or book-to-market equity.

Constructing portfolios sorted by book-to-market equity

1. Sort stocks into 10 portfolios based on book-to-market equity (i.e., cutoff at each 10th percentile).
2. Compute returns on these portfolios over the subsequent year.
3. Rebalance annually.

Cumulative returns on low vs. high book-to-market equity

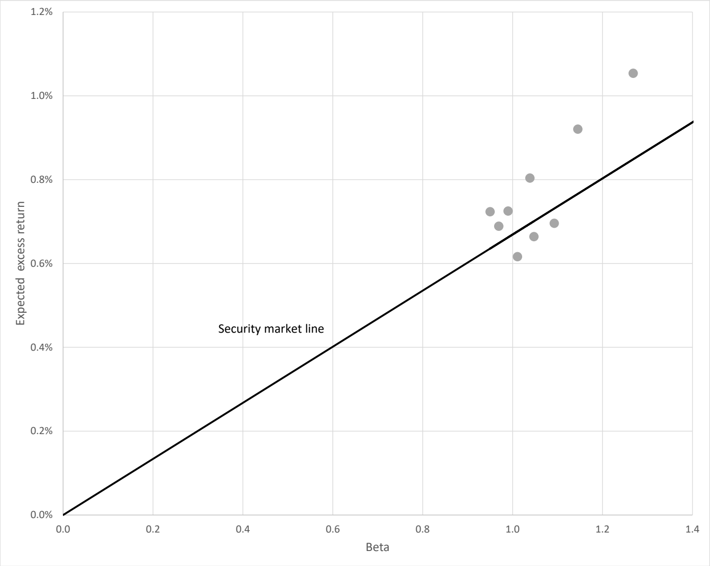
Cumulative returns on low vs. high book-to-market equity (1928–1938)

Cumulative returns on low vs. high book-to-market equity (2006–2016)

Example 3: Portfolios sorted by book-to-market equity

	Portfolio									
Statistic	Low	2	3	4	5	6	7	8	9	High
Mean (%)	0.62	0.72	0.69	0.66	0.73	0.80	0.70	0.92	1.05	1.05
SD (%)	5.68	5.30	5.39	5.90	5.60	6.03	6.39	6.74	7.63	9.12
Sharpe ratio	0.11	0.14	0.13	0.11	0.13	0.13	0.11	0.14	0.14	0.12
Market beta	1.01	0.95	0.97	1.05	0.99	1.04	1.09	1.15	1.27	1.45
Alpha (%)	−0.06	0.09	0.04	−0.04	0.06	0.11	−0.04	0.15	0.21	0.08
Alpha (*t*-stat)	−1.14	1.92	0.92	−0.67	1.15	1.55	−0.45	1.84	1.97	0.57

Alpha on portfolios sorted by book-to-market equity



Summary

- Market-timing strategy: Increase allocation to stocks relative to the riskless asset when the price-earnings ratio is low.
- Value strategy: Invest in value stocks that are trading at high book-to-market equity.

Next steps

- Multi-factor models that generalize the CAPM and allow more sources of systematic risk such as value to be priced.
- Use the multi-factor model for performance evaluation and event studies.